Yihong Zhu

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Education

Stanford University June 2025 - Present

Ph.D. Aeronautics & Astronautics **GPA: 4.0/4.0**

Research Area: Computational Mechanics (Fluids/Structures), Physics-based Surrogate Modeling

Advisor: Charbel Farhat (2026 Guggenheim Medal Winner, US & UK National Academy of Engineering)

University of California, San Diego September 2021 - June 2025

B.S. Aerospace Engineering, Minor in Mathematics **GPA: 3.967/4.000**

Publications

- C. Farhat, **Y. Zhu**, F. As'ad, T. Taddei; The Snapping Phenomenon and Boundary Enforcement in Model Reduction: Mechanisms and Mitigation (under review at CMAME)
- **Y. Zhu**, S. Sato, O. Schmidt; SPOD Mode Interpolation with Frequency Alignment on the Grassmannian Manifold. *AIAA SCITECH 2026 Forum. 2026*
- Y. Weng, **Y. Zhu**, A. Saha; Vortex-driven subharmonic bifurcation in a multi-flame Rijke tube. *Combustion and Flame, 282, p.114447, 2025*
- Y. Weng, **Y. Zhu**, V. Unni, R. Sujith, A. Saha; Deriving Early Warning Signals for Turbulent Thermoacoustic Systems using Principal Component Analysis. (In preparation)
- **Y. Zhu**, H. Sharma; Hamiltonian Adaptive Variational Integrators for Nonconservative Problems in Astrodynamics. *AIAA ARC 2024 Regional Student Conferences, p. 80056*

Conferences

- Orlando, FL. *AIAA SCITECH 2026 Forum* January 12 - 16th, 2026
  Y. Zhu, S. Sato, O. Schmidt; SPOD Mode Interpolation with Frequency Alignment on the Grassmannian Manifold
- Salt Lake City, UT. *77th Annual Meeting of the APS Division of Fluid Dynamics* November 24 - 26th, 2024
  Y. Weng, **Y. Zhu**, V. Unni, R. Sujith, A. Saha; Deriving early warning signals of turbulent thermoacoustic system using principal component analysis
- Salt Lake City, UT. *77th Annual Meeting of the APS Division of Fluid Dynamics* November 24 - 26th, 2024
  Y. Weng, **Y. Zhu**, A. Saha; Flame-driven subharmonic bifurcation in a multi-flame Rijke tube
- Irvine, CA. 2024 *SoCal Fluids XVII* April 13th, 2024
  Y. Weng, **Y. Zhu**, V. Unni, R. Sujith, A. Saha; Detecting Early Warning Signals of Turbulent Thermoacoustic Systems Using Principal Component Analysis
- Santa Clara, CA. *2024 AIAA Region VI Student Conference* March 23 - 24th, 2024
  Y. Zhu, H. Sharma; Hamiltonian Adaptive Variational Integrators for Nonconservative Problems in Astrodynamics
- Northridge, CA. *Western States Section Combustion Institute 2023 Fall Technical Meeting* October 16 - 17th, 2023
  Y. Weng, **Y. Zhu**, A. Saha; Asymmetric Couplings in a Multi-flame Laminar Rijke Tube

Technical Experience

Rocket Propulsion Laboratory (RPL) | Student Organization, UCSD October 2021 - April 2025

Propulsion Lead Engineer April 2023 - April 2025

- Composed documents on nozzle design, combustion, and rocket stability dynamics for future training and reference.
- Performed hydrostatic testing on propellant tanks to ensure their structural integrity during flight.
- Modeled the fuel combustion process using thermochemical theories to obtain engine performance parameters.

Propulsion Engineer June 2022 - April 2023

- Designed airfoil fins using Ansys Fluent to reduce drag and ensure rocket stability throughout the ascending phase.
- Derived a rocket stability ODE model to show our rocket remains within ±2° AoA under gusts using MATLAB *ode45*.

New Member Project (Solid Rocket) Engineer October 2021 - April 2022

- Designed and built a 1-meter-long model rocket that reaches about 1400m altitude with OpenRocket Simulations.
- Incorporated Arduino as the flight computer to record altitude and velocity for parachute deployment purposes.

Course Projects | UCSD & Stanford

Graduate Course AA214: Numerical Methods for Compressible Flows, Stanford March 2026

- Implemented a 2D Finite Volume Compressible Euler Solver on an unstructured grid with MATLAB to solve a supersonic ramp intake problem.

Undergraduate Course MAE185: Computational Fluid Mechanics, UCSD March 2024

- Implemented Direct Numerical Simulation (DNS) for the 2D viscous Navier-Stokes equation using finite differences in MATLAB for Mach 4 flow over a flat plate.

Research Experience

Research Assistant | Farhat Research Group, Stanford University June 2025 - Present

Advisor: Prof. Charbel Farhat

- Implemented a boundary-condition-enforcing shift vector for reduced-order models (ROMs) in AERO-F (our in-house C++ CFD/ROM code) that prevents catastrophic failure modes called “snapping” in ROMs.
- Developed a reduced-order dynamic substructuring Python codebase for studying scalable substructuring methods in structural dynamics.

Research Assistant | Computational Modeling and Flow Physics, UCSD August 2024 - June 2025

Advisor: Prof. Oliver Schmidt

- Developed Grassmannian interpolation schemes for spectral proper orthogonal decomposition reduced-order basis.

Research Assistant | Flame-Fluids-Energy Lab, UCSD June 2023 - August 2024

Advisor: Prof. Abhishek Saha

- Researched thermoacoustic instability and flame dynamics using Rijke tube experiments with Mie-scattering imaging.
- Applied statistical analysis using MATLAB to describe flame-flame interactions in a multi-hole burner setup.
- Used data-driven techniques like Principal Component Analysis (PCA) to quantify variations in phase space topology.

Research Assistant | Numerical Integrators, UCSD September 2022 - August 2023

Advisor: Dr. Harsh Sharma (Professor at UW-Madison since August 2025)

- Studied dynamical systems and Hamiltonian mechanics, and their applications to Geometric Numerical Integrators.
- Derived Hamiltonian Adaptive Variational Integrator for nonconservative systems (e.g., Low Earth Orbit with drag), and showed its symplecticity, time-adaptivity, and energy preservation.

Teaching Experience

Course Grader | UCSD Department of Mechanical and Aerospace Engineering September 2023 - June 2025

- MAE 101A - Introductory Fluid Mechanics (Winter 2025)
- MAE 104 - Aerodynamics (Fall 2024)
- MAE 101B - Advanced Fluid Mechanics (Winter, Spring 2024)
- MAE 11 - Thermodynamics (Fall 2023)

Course Tutor | UCSD Academic Achievement Hub September 2022 - December 2022

- MATH 4C - Precalculus for Science and Engineering (Fall 2022)

Skills/Interests

Programming/Computation: MATLAB, Python, C++, Linux, familiar with HPC clusters

Software: Solidworks CAD, Solidworks FEA, Ansys Fluent, Ansys FEA

Misc: 3D Printing